Buried in Textual Debt: Context Pruning with Visual Evidence Preservation for MLLM Agents

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Abstract

Multimodal Large Language Models (MLLMs) are increasingly deployed as multi-step agents, where explicit reasoning supports task decomposition and tool coordination but also accumulates self-generated text. Over long trajectories, this text can dominate the context and suppress visual evidence, creating *textual debt*. We observe that reasoning becomes redundant once task-relevant visual evidence is grounded, while stale hypotheses can misguide later inference when grounding remains uncertain. Pruning must therefore remove redundant text without discarding visual evidence. We propose SPARE, a Kullback–Leibler (KL)-guided framework for pruning accumulated reasoning in multimodal tool-use agents. SPARE uses a compact task-state summary as privileged diagnostic context. For each candidate segment, it replays the same model under the original and summary-conditioned contexts. Reverse-KL divergence from on-policy self-distillation (OPSD) then tests whether the summary sufficiently covers the segment without disrupting future reasoning. We further fine-tune the summarizer with supervised fine-tuning (SFT), enabling more compact summaries, broader coverage, and more aggressive pruning. Across multi-step visual tool-use benchmarks, SPARE achieves the highest average accuracy among pruning methods while removing 37.89–64.58% of reasoning tokens. This favorable accuracy–context trade-off shows that reducing textual dominance restores reliance on visual evidence and mitigates over-conditioning on self-generated language.

1 Introduction

Multimodal Large Language Models (MLLMs) are increasingly deployed as agents that reason, call tools, inspect intermediate observations, and revise their answers over time [1]. Explicit reasoning is beneficial in this setting, as it helps agents decompose visual tasks, coordinate multi-step tool use, interpret returned observations, and maintain progress across turns [2]. Yet it also introduces a largely overlooked cost: each interaction step adds self-generated text to the working context. Over long trajectories, this accumulated text can dominate computation relative to the original image and newly returned observations, making the model increasingly conditioned on its own plans, descriptions, and assumptions rather than external visual evidence.

Not all reasoning history remains useful throughout a multimodal reasoning trajectory. When early reasoning correctly captures task-relevant visual evidence, the accumulated text may still provide a faithful abstraction of the image. However, when early reasoning fails to attend to the relevant visual regions, the model may elaborate on an incomplete or weakly grounded textual state. In this case, additional reasoning tokens become *textual debt*: stale linguistic context that contributes little new evidence while competing with image tokens for attention and context budget. From this view, pruning reasoning tokens is not merely a way to reduce computation or sequence length. It can

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also function as a grounding mechanism by reducing over-conditioning on stale textual context and restoring the opportunity to re-attend to the visual input.

Existing approaches do not directly address this problem. Visual-token compression reduces redundancy in the *visual* stream by dropping or merging image tokens [3, 4, 5, 6, 7]; this is effective for compute, but it targets a modality that is often already under-attended in deep layers, and compressing it further can weaken the relative contribution of image evidence [8]. Concise-reasoning methods primarily shorten newly generated rationales, and generic summarization compresses history at a coarse level, but neither diagnoses which historical segments remain functionally necessary. In multi-step agents, the more pressing source of redundancy is the self-generated text that accumulates across rounds.

We therefore reframe context compression as a light-weight, *evidence-preserving selective forgetting* problem and instantiate it as **SPARE** (Selective Pruning of Accumulated Reasoning with Visual Evidence Preservation), a KL-guided reasoning-pruning framework for multimodal tool-use agents. The key idea is to use a compact task-state summary as *privileged diagnostic context* rather than as a replacement for the history: instead of overwriting the trajectory with the summary, we test whether each historical segment retains information beyond the consolidated state. Concretely, SPARE replays the same model under the original and summary-conditioned contexts and uses reverse-KL between token-level continuation distributions to measure residual dependence on the original segment. Segments with low residual sensitivity are pruned, while high-sensitivity segments are preserved as evidence-critical content. This diagnostic requires neither an external verifier nor a separate reward model. We further fine-tune the summarizer with supervised fine-tuning (SFT), enabling more compact summaries, broader coverage, and more aggressive pruning. Across multi-step visual tool-use benchmarks, SPARE achieves the highest average accuracy among pruning methods while removing 37.89–64.58% of reasoning tokens. We also show that suppressing textual dominance increases attention to image tokens, supporting our claim that reducing textual debt restores reliance on visual evidence.

Our contributions are summarized as follows:

- We propose SPARE, an evidence-preserving selective-forgetting method that uses summary-conditioned KL divergence to estimate the functional redundancy of prior reasoning and prune segments whose information has already been consolidated.
- We further fine-tune the summarizer with SFT to produce more compact task-state summaries, enabling broader coverage and more aggressive pruning while preserving task-relevant information.
- Experiments across multiple backbones and visual tool-use benchmarks show that SPARE improves the accuracy and restores the influence of visual evidence on downstream reasoning.

2 Related Work

Token Pruning for MLLM Existing MLLM token pruning methods primarily aim to remove redundancy from the visual stream [3, 4, 5, 7, 9, 10]. While effective for efficiency, this objective can be misaligned with multimodal reasoning tasks, where many questions are partially answerable from textual cues and visual evidence is already under-attended in deep layers [11, 12, 13, 8]. Concise-reasoning and generic summarization methods can shorten generated text or compress history, but they do not directly diagnose which historical reasoning segments remain functionally necessary. Our method instead targets accumulated *textual* redundancy in multi-step tool-use trajectories and prunes only segments already covered by a compact task-state summary.

Mitigating Language-Prior Bias A related line of work mitigates language-prior bias or weak visual grounding in multimodal models. Prior approaches include counterfactual training for VQA [14], contrastive decoding [15], attention calibration [16, 17, 18], and preference optimization for image-grounded responses [19, 20]. These methods shift attention toward visual evidence through training, decoding, or attention reweighting, but they leave the competing textual redundancy intact and often require extra inference or auxiliary data. In contrast, our method operates at test time, requires no auxiliary model, and improves grounding by pruning stale self-generated text that competes with visual evidence.

3 Preliminaries

3.1 Multi-Step Tool Use in MLLMs

We consider a vision language model (VLM) π_θ that answers a user question through multi-step, tool-augmented reasoning. The input consists of a question q and a set of image tokens $\mathcal{I}$, and the model may query a fixed set of external tools $\mathcal{T}$ over at most T interaction steps.

At step t, the model conditions on the current message history $\mathbf{H}_t$ and produces an assistant message

$$m_t = (h_t, a_t), \tag{1}$$

where h_t is a free-form reasoning span and a_t is either a tool invocation $\langle$tool_call$\rangle$ or a final answer $\langle$response$\rangle$. If a_t invokes a tool, the corresponding tool result is appended to the history, yielding $\mathbf{H}_{t+1}$, and the interaction proceeds to the next step; if a_t emits a final response, the episode terminates. We denote the sequence of past reasoning–action segments by $\mathcal{S}_t = \{\, s_i = (h_i, a_i) \,\}_{i<t}$.

3.2 On-policy Self-Distillation

Knowledge distillation trains a student model π_θ to match a teacher distribution π_{teacher} over next tokens. Conventional off-policy distillation applies this objective on fixed teacher or ground-truth sequences, which creates exposure bias because the training prefixes differ from the student-generated prefixes encountered at inference [21]. *On-policy* distillation reduces this mismatch by applying the distillation loss to trajectories sampled from the student itself, so that the model is supervised on its own inference-time states [21].

In the on-policy self-distillation instantiation, the teacher and student share the same model, so no external teacher is required [22]. Given a multimodal input $(q, \mathcal{I})$, we first sample a response $y \sim \pi_\theta(\cdot \mid q, \mathcal{I})$ and then align the student's per-token distribution with a teacher distribution defined over the same model:

$$\mathcal{L}_{\text{OPSD}}(\theta) = \mathbb{E}_{(q,\mathcal{I})}\, \mathbb{E}_{y\sim\pi_\theta(\cdot|q,\mathcal{I})} \left[\frac{1}{L} \sum_{k=1}^{L} D_{\text{KL}}\Big(\pi_{\text{teacher}}(\cdot \mid y_{<k}, q, \mathcal{I}) \,\Big\|\, \pi_\theta(\cdot \mid y_{<k}, q, \mathcal{I}) \Big) \right]. \tag{2}$$

Here the gradient is taken with respect to the student parameters θ, while the teacher distribution is treated as a fixed target. Sampling y from π_θ makes the objective on-policy, and defining π_{teacher} from the same model makes it self-distillation.

Our method adopts this OPSD view only as a diagnostic principle: rather than optimizing a distillation objective, we replay the same model under two related contexts and use the resulting distributional shift to estimate whether a historical reasoning segment still contains information not covered by a compact task-state summary.

Attention over modalities. For visualization, we also measure text-to-image attention across decoder layers following prior work [3]; the exact definition is provided in Appendix A.

4 Method

We propose **SPARE** (Selective Pruning of Accumulated Reasoning with Visual Evidence Preservation), a post-hoc context-pruning method for multi-step MLLM agents. The motivation is that explicit reasoning is useful for decomposing visual questions, planning tool calls, and integrating intermediate observations, but retaining the entire self-generated textual trace indefinitely can create *textual debt*: the agent's later predictions become increasingly conditioned on its own textual history rather than on the visual evidence underlying the task. SPARE therefore does not aim to suppress reasoning generation. Instead, it diagnoses which accumulated text has low residual sensitivity to a compact task state and compresses only those low-residue segments, while preserving modality-critical content such as OCR strings, coordinates, bounding boxes, visual anchors, tool calls, and external observations. In this sense, SPARE compresses linguistic scaffolding rather than task state.

4.1 Summary-Conditioned Coverage Estimation

Adaptive summary trigger. SPARE is invoked only after the student agent itself calls an internal `summarize_the_task` tool and produces a compact task-state summary σ. This self-generated

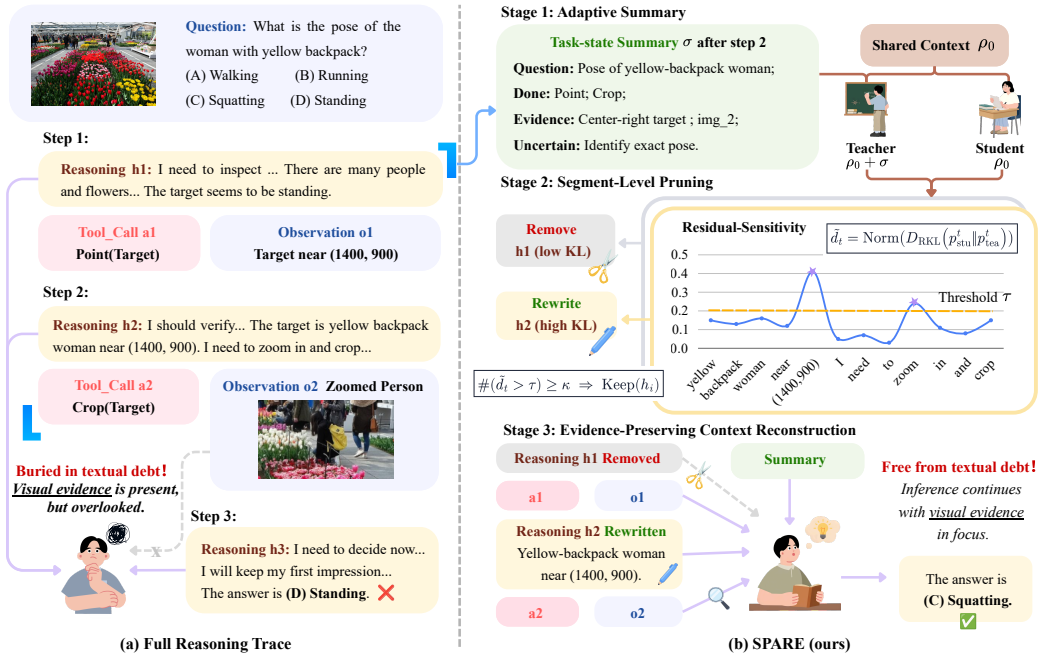


Figure 1: Overview of SPARE. (a) In the full reasoning trace, accumulated self-generated text reinforces a stale hypothesis and obscures visual evidence returned by the tools, leading to an incorrect answer. (b) SPARE invokes a compact task-state summary as a *transient probe* that is not written into the persistent trajectory. It replays each historical reasoning segment with and without summary conditioning and measures its token-level residual sensitivity using normalized reverse KL. A count-based rule prunes summary-covered segments with insufficient high-KL residue while retaining evidence-critical segments. Each pruned segment is replaced by concise structured visual evidence extracted from its original reasoning, whereas planning-dominated text is removed and the original tool calls, observations, and images remain unchanged. The resulting context reduces textual debt, refocuses subsequent inference on visual evidence, and produces the correct answer.

summary contain the original question, completed tool calls, accumulated visual evidence, and remaining uncertainty. We use σ only as auxiliary diagnostic context to test which prior reasoning spans are already covered by the consolidated task state. This adaptive trigger avoids fixed-interval compression: short trajectories incur no pruning cost, and longer trajectories are considered for pruning only after the student has explicitly produced a compact state.

Summary-conditioned replay. Let $\boldsymbol{x} = (x_1, \ldots, x_N)$ be the tokenized concatenation of candidate reasoning segments, separated by a fixed delimiter. To test whether the self-generated summary σ covers a segment's information, we replay $\boldsymbol{x}$ under two contexts:

$$\rho_{\text{stu}} = \rho_0, \tag{3}$$

$$\rho_{\text{tea}} = \rho_0 \,\|\, \sigma, \tag{4}$$

where ρ_0 contains the shared system, tool-use, and task prefix. The teacher context receives σ as privileged information, while the student context does not. Importantly, this does not introduce an external teacher model: the teacher distribution is produced by the same model, conditioned on the self-generated summary. The same model π_θ is then queried under both contexts:

$$p_k^{\text{stu}}(u) = P_{\pi_\theta}(u \mid \rho_{\text{stu}}, \boldsymbol{x}_{<k}), \tag{5}$$

$$p_k^{\text{tea}}(u) = P_{\pi_\theta}(u \mid \rho_{\text{tea}}, \boldsymbol{x}_{<k}), \tag{6}$$

where k indexes replay tokens and u indexes vocabulary tokens. If adding σ changes the continuation distribution only slightly, the summary already covers the replayed content; if it induces a large shift, the original reasoning contains information not captured by the summary and should be preserved.

Top-K reverse-KL coverage score. For each replay token position k, we compute a truncated reverse-KL score over the student's top-K vocabulary support Ω_k^{stu}:

$$d_k = \left[\sum_{u \in \Omega_k^{\mathrm{stu}}} p_k^{\mathrm{stu}}(u) \left(\log p_k^{\mathrm{stu}}(u) - \log p_k^{\mathrm{tea}}(u) \right) \right]. \tag{7}$$

We use $K = 20$ in all experiments. If $u \in \Omega_k^{\mathrm{stu}}$ is absent from the teacher top-K support, $\log p_k^{\mathrm{tea}}(u)$ is set to -20. The resulting score d_k measures the residual information not covered by the summary: high values indicate that the token remains summary-sensitive and should be protected, while low values indicate that the summary sufficiently explains the token's context.

KL scales vary across pruning events, we normalize scores within each event:

$$\tilde{d}_k = \frac{d_k - d_{\min}}{d_{\max} - d_{\min} + \epsilon}, d_{\min} = \min_{k'} d_{k'}, d_{\max} = \max_{k'} d_{k'}. \tag{8}$$

We set $\epsilon = 10^{-12}$ and assign all normalized scores to zero when $d_{\max} = d_{\min}$. Segment-level pruning then aggregates $\tilde{d}_k$ to determine whether the summary covers each reasoning segment.

4.2 Segment-Level Pruning

Segment token sets. The replay sequence $\boldsymbol{x}$ is formed by concatenating historical assistant reasoning segments. Let h_i be the i-th candidate segment and

$$\mathcal{P}_i = \{k : x_k \in h_i\} \tag{9}$$

denote its token positions in the replay sequence. Since segment boundaries are known during concatenation, token-level residual-sensitivity scores can be mapped back to their original segments.

Keep rule. A segment is retained if it contains at least κ high-sensitivity tokens:

$$\mathrm{Keep}(h_i) = \mathbf{1}\left\{ \left| \left\{ k \in \mathcal{P}_i : \tilde{d}_k > \tau \right\} \right| \geq \kappa \right\}. \tag{10}$$

High-KL tokens indicate residual textual influence not covered by the compact task-state summary. Thus, segments with enough high-KL residue are kept verbatim, while low-residue segments are removed.

Why a count-based threshold. We use a count rule rather than an average score because modality-critical evidence is often sparse. A long reasoning segment may be mostly obsolete but still contain a few crucial tokens, such as an OCR string, coordinate, bounding box, or candidate label. Averaging can dilute these signals, whereas the count rule conservatively preserves segments containing even sparse high-sensitivity evidence.

4.3 Evidence-Preserving Pruning

Summary as auxiliary context. SPARE does not rewrite the assistant trace or insert the summary into it. The compact task-state summary is used only as auxiliary context for the next reasoning step, matching our controlled baselines and avoiding rewriting artifacts. A large KL divergence suggests that self-generated summaries do not faithfully preserve prior tool-use reasoning, so inserting them into the context may distort the original reasoning state.

KL-driven reconstruction. After reverse-KL scoring, pruning is applied at the segment level: high-KL segments are preserved by converting lengthy reasoning into concise, verifiable visual evidence, such as crop locations, recognized text, numeric values, or relative relations among candidates. Low-KL segments are dropped because their information is already captured by the summary; in those steps, the assistant reasoning text is removed entirely, while tool outputs, visual observations, and image tokens remain unchanged. High-KL segments indicate residual reasoning that is not sufficiently recoverable from the summarized history of previous tool interactions.

Table 1: Task performance and average reasoning pruning across five visual tool-use benchmarks. Benchmark columns report accuracy. Acc. and Pr. denote macro-average accuracy and reasoning-token pruning rate, respectively. Note that MNMS evaluates tool planning and therefore is inapplicable for Direct Answer mode.

Model / Metric	GTA ↑	V* ↑	B-Jig. ↑	VTB ↑	MNMS ↑	Acc. ↑	Pr. %
Qwen3-VL-30B-A3B-Instruct							
Tool Baseline (Full Trace)	45.76	55.85	67.33	**27.31**	36.84	44.30	0.00
− Tools (Direct Answer)	33.90	52.36	**70.67**	21.43	–	43.42	0.00
+ Delete All Reasoning	50.85	37.17	57.33	26.05	24.56	35.22	99.86
+ Visual Evidence-Only	**59.32**	61.78	64.00	26.47	25.44	42.73	77.09
+ **SPARE (ours)**	**59.32**	**62.30**	64.00	26.47	**49.56**	**49.19**	63.70
Qwen3-VL-8B-Instruct							
Tool Baseline (Full Trace)	42.37	72.77	**74.00**	23.53	**60.96**	**54.27**	0.00
− Tools (Direct Answer)	42.37	63.35	72.00	20.17	–	47.34	0.00
+ Delete All Reasoning	40.68	68.59	62.00	23.53	57.02	50.12	100.00
+ Visual Evidence-Only	44.07	69.63	59.33	**24.79**	56.58	50.35	72.58
+ **SPARE (ours)**	**49.15**	**73.30**	72.00	23.11	57.46	53.47	37.89
Llama-3.1-Nemotron-Nano-VL-8B-V1							
Tool Baseline (Full Trace)	40.68	48.69	45.33	6.16	**26.75**	30.10	0.00
− Tools (Direct Answer)	**44.07**	**56.02**	**51.33**	**9.46**	–	**36.44**	0.00
+ Delete All Reasoning	37.29	45.03	46.67	6.75	21.49	28.07	99.38
+ Visual Evidence-Only	37.29	43.98	44.00	6.26	17.98	26.32	54.84
+ **SPARE (ours)**	42.37	46.60	48.67	**9.46**	21.93	29.97	64.58

4.4 Strengthening the Summarizer via SFT

The effectiveness of SPARE depends on the quality of the task-state summary: a summary that covers more of the accumulated reasoning enables more aggressive low-residue pruning without disrupting future decisions. To this end, we further fine-tune the summarizer via supervised fine-tuning (SFT). A stronger summarizer expands the coverage of each summary, so that a larger fraction of reasoning segments can be safely explained by the summary alone. Under the same reverse-KL criterion, more segments therefore fall into the low-residue regime and become prunable, while high-residue visually-grounded segments remain preserved.

5 Experiments

5.1 Experimental Settings

Benchmarks. We evaluate SPARE on several visual tool-use benchmarks requiring multi-step reasoning, tool invocation, and intermediate visual observations. **VisualToolBench** (VTB) [23] requires models to actively "think with images" through operations such as cropping, editing, and enhancement. **m&m's** (MNMS) [24] includes 4000+ multi-step multimodal tasks over 33 tools, covering multimodal models, public APIs, and image-processing modules. **GTA** [25] provides real-world tasks with human-written queries and executable tool chains across perception, operation, logic, and creativity. **V*** [26] evaluates guided visual search on high-resolution images, requiring fine-grained target localization before answering. **BLINK-Jigsaw** (B-Jig.) [27] tests spatial perception by requiring models to reassemble image fragments. Together, these benchmarks cover long-horizon tool use and perception-intensive reasoning, making them suitable for testing whether reasoning compression preserves visual grounding.

Backbones. To evaluate SPARE's generality, we test three vision–language backbones: Qwen3-VL-8B-Instruct, Qwen3-VL-30B-A3B-Instruct, and Llama-3.1-Nemotron-Nano-VL-8B-V1. We keep generation settings and the tool interface fixed.

Compared methods. For each backbone, we compare five inference strategies under an identical tool-use harness. (i) **Tool Baseline (Full Trace)** executes the standard multi-round tool loop and retains the complete reasoning–action–observation history. (ii) **- Tools (Direct Answer)** answers

the query directly without invoking tools and serves as a non-agentic reference. (iii) **+ Delete All Reasoning** removes every completed reasoning span while preserving the original action blocks, tool observations, and images. (iv) **+ Visual Evidence-Only** non-selectively rewrites every completed reasoning span as compact structured visual evidence while preserving its original action block. (v) **+ SPARE (Ours)** first uses the task-state summary as a probe, estimates the residual sensitivity of historical reasoning through summary-conditioned reverse KL, and reconstructs only selected low-residue segments as structured visual evidence. If the probe selects no segment for pruning, the summary is discarded and does not enter the subsequent history; if pruning is triggered, the summary interaction may be retained together with the reconstructed context. Full Trace represents complete context retention, whereas Delete All Reasoning and Visual Evidence-Only provide non-selective compression controls for evaluating SPARE's segment selection and evidence-preserving reconstruction.

Metrics. Following each benchmark's official protocol, we report task success or accuracy, where higher is better (↑). We also report **Pruned (%)**, the net percentage of reasoning-side history tokens removed relative to Tool Baseline. Token counts include remaining reasoning, structured evidence blocks, and any task-state summary, but exclude unchanged action blocks, tool observations, and image tokens. Thus, Full Trace has 0.0% pruning, Delete All Reasoning has 100.0%, and Direct Answer is assigned 0.0% for reporting consistency as it produces no comparable trajectory. The main table reports the macro-average pruning ratio across benchmarks on the common evaluation set.

Implementation details. SPARE is applied purely at test time. Pruning is triggered only when the model invokes the internal `summarize_the_task` tool, so short trajectories without summaries incur no compression cost. We compute the truncated reverse-KL coverage score on the student's top-K support and map token scores to reasoning segments using the count-based rule with fixed $\tau = 0.2$ and $\kappa = 2$ across all models and benchmarks (Section 4). The same model provides both student and teacher distributions by conditioning on prompts with and without the task-state summary σ, avoiding any auxiliary model.

5.2 Main Results

Table 1 reports task accuracy and reasoning pruning across five benchmarks. Across all three backbones, SPARE consistently achieves the highest average accuracy among the pruning methods, while matching or outperforming Full Trace overall. On Qwen3-VL-30B, it improves average accuracy from 44.30% to 49.19% while pruning 63.70% of the reasoning tokens. On Qwen3-VL-8B, it preserves comparable performance (53.47% vs. 54.27%) with 37.89% pruning. Nemotron shows a different pattern: Direct Answer outperforms Full Trace on all applicable benchmarks, consistent with weaker tool-use capability causing the accumulated tool-use context to interfere with later decisions. Even so, SPARE improves Full Trace on GTA, B-Jigsaw, and VTB and nearly matches its overall accuracy under 64.58% pruning. This suggests that selective context reduction can partially mitigate tool-induced textual debt without discarding the benefits of the visual interaction history. Additional analyses further support the textual-debt hypothesis: a controlled diagnostic shows that SPARE is less likely to follow stale textual hypotheses than both Full Trace and random deletion, while attention visualizations show increased reliance on image tokens after pruning (Appendix B).

5.3 SFT for Summarization Capability

We further study whether stronger task-state summaries enable more aggressive pruning. To this end, we collect 376 on-policy trajectories from Qwen3-VL-8B on VTC-Bench (VTC) [28] and prompt the stronger Qwen3-235B to produce a compact task-state summary at each intermediate context. Qwen3-VL-8B is then fine-tuned on the resulting (context, summary) pairs with a standard next-token objective, distilling the 235B model's summarization capability into the 8B agent. Table 2 reports task accuracy under the SPARE pipeline before and after SFT-based summarizer fine-tuning; the last row shows the relative increase in

Table 2: SFT results on VTB and MNMS.

Model / Metric	**VTB**	**MNMS**
Base (Qwen3-VL-8B)	26.83	60.61
+ SFT summarizer	**28.73**	**65.44**
More pruned tokens vs. Base	**23.8%**	**13.5%**

pruned tokens per trajectory compared with the base summarizer. The SFT summarizer enables more aggressive pruning and improves task accuracy on both VisualToolBench and MNMS, supporting our motivation that stronger summaries expand the safe-to-prune region and allow more redundant reasoning to be removed without harming downstream decisions.

5.4 Ablation Study

Why combine selective pruning with evidence reconstruction? We conduct ablations using Llama-3.1-Nemotron-Nano-VL-8B-V1 on GTA, with the results summarized in Table 3. **Full trace** retains all reasoning as the reference. **Summary probe only** summarizes the current task state to assist subsequent reasoning but does not prune the history. **+ KL selection** deletes low-KL segments while retaining high-KL segments in their original form. **+ Evidence** omits KL selection and replaces every reasoning segment with structured visual evidence. The complete SPARE removes low-KL segments and reconstructs high-KL segments as concise visual evidence. The results show that summary assistance alone is insufficient. KL selection avoids unsafe pruning but provides no compression when high-KL reasoning is retained verbatim, whereas evidence-only reconstruction compresses the history at the cost of accuracy. Combining selection and reconstruction achieves the best accuracy (42.37%) with substantial pruning (62.42%), demonstrating that the two components are complementary.

Table 3: Ablations on GTA. Accuracy and pruning rates are percentages.

Ablation Setting	Acc.	Pr.
Selective Pruning and Evidence Reconstruction		
Full trace baseline	40.68	0.00
Summary probe only	35.59	0.00
+ KL selection	40.68	0.00
+ Evidence	37.29	56.08
+ KL selection + Evidence (ours)	**42.37**	**62.42**
Adaptive Summary Triggering		
Every round	32.20	52.54
Before final answer	38.98	51.38
Model-selected (ours)	**42.37**	**62.42**
Parameter Robustness ($\tau = 0.2$)		
$\kappa = 1$	**42.37**	62.42
$\boldsymbol{\kappa = 2}$ **(default)**	**42.37**	62.42
$\kappa = 3$	38.98	**63.16**

When should the context be summarized and pruned? The second block compares three triggering policies. **Every round** forces the model to summarize after each reasoning round, whereas **Before final answer** invokes the summary once immediately before answering. **Model-selected** allows the model to decide whether and when summarization is necessary. Frequent forced summarization can disrupt reasoning, while summarizing only before the final answer cannot control earlier context accumulation. The model-selected policy achieves the highest accuracy (42.37%) and pruning rate (62.42%), demonstrating that adaptive summarization is more effective than fixed triggering schedules. We next ablate the KL-based decision rule by varying the required number of consecutive pruning decisions, $\kappa \in 1, 2, 3$, while fixing $\tau = 0.2$. The settings $\kappa = 1$ and $\kappa = 2$ yield identical results. Increasing κ to 3 slightly improves the pruning rate but reduces accuracy. Based on this local stability, we adopt $(\tau, \kappa) = (0.2, 2)$ as the default setting.

6 Conclusion

In this paper, we identify *textual debt* as a key failure mode of multi-step MLLM agents, where accumulated self-generated reasoning gradually dominates the context and weakens reliance on visual evidence. Our motivation is that pruning reasoning tokens can help because their utility changes over time: once task-relevant visual information has been captured, later text often becomes redundant, while incorrect or incomplete early grounding can make accumulated text reinforce stale linguistic assumptions. Based on this insight, we propose SPARE, a OPSD KL-guided framework that selectively removes redundant reasoning while preserving visual evidence, thereby reducing textual dominance and redirecting attention to images. Experiments show that SPARE improves task performance, reduces reasoning-token usage, and restores attention to visual evidence, suggesting that inference-time selective forgetting is an effective mechanism for long-context multimodal reasoning.

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A Additional Preliminaries

A.1 Multi-Step Tool Use in MLLMs

We consider a vision language model (VLM) π_θ that answers a user question through multi-step, tool-augmented reasoning. The input consists of a question q and a set of image tokens $\mathcal{I}$, and the model may query a fixed set of external tools $\mathcal{T}$ over at most T interaction steps.

At step t, the model conditions on the current message history $\mathbf{H}_t$ and produces an assistant message

$$m_t = (h_t, a_t), \tag{11}$$

which is composed of a free-form reasoning span h_t and an action block a_t. The action block is either a tool invocation $\langle$tool_call$\rangle$ or a final answer $\langle$response$\rangle$. If a_t invokes a tool, the corresponding tool result is appended to the history, yielding $\mathbf{H}_{t+1}$, and the interaction proceeds to the next step; if a_t emits a final response, the episode terminates. We denote the sequence of past reasoning–action segments by $\mathcal{S}_t = \{\, s_i = (h_i, a_i) \,\}_{i<t}$.

A.2 On-policy self-distillation.

Knowledge distillation trains a student model π_θ to match a teacher distribution π_{teacher} over next tokens. Conventional off-policy distillation minimizes this divergence on fixed teacher or ground-truth sequences, which creates exposure bias because training prefixes differ from the student-generated prefixes encountered at inference [21]. *On-policy* distillation reduces this mismatch by applying the distillation loss to trajectories sampled from the student itself, so the model is supervised on its own inference-time states [21].

In the on-policy self-distillation instantiation, the teacher and the student share the same model, so no external teacher is required [22]. Concretely, given a multimodal input $(q, \mathcal{I})$, we first roll out a response on-policy, $y \sim \pi_\theta(\cdot \mid q, \mathcal{I})$, and then align the student's per-token distribution with a teacher distribution π_{teacher} defined over the same model:

$$\begin{aligned}\mathcal{L}_{\text{OPSD}}(\theta) = \mathbb{E}_{(q,\mathcal{I})}\, \mathbb{E}_{y\sim\pi_\theta(\cdot|q,\mathcal{I})} \Bigg[\frac{1}{L} \sum_{k=1}^{L} \\ D_{\text{KL}}\Big(\pi_{\text{teacher}}(\cdot \mid y_{<k}, q, \mathcal{I}) \,\Big\|\, \pi_\theta(\cdot \mid y_{<k}, q, \mathcal{I})\Big)\Bigg],\end{aligned} \tag{12}$$

where the gradient is taken with respect to the student parameters θ and the teacher distribution is treated as a fixed target (stop-gradient). Sampling y from π_θ makes the objective on-policy, while defining π_{teacher} from the same model makes it self-distillation. This formulation lets the model learn from its own generated trajectories and provides the training signal on which our method builds.

A.3 Self-attention over modalities.

Within a transformer layer ℓ with attention heads indexed by b, the attention weight from a query token i to a key token j is

$$A_{ij}^{(\ell,b)} = \text{softmax}_j\left(\frac{q_i^{(\ell,b)} \cdot k_j^{(\ell,b)}}{\sqrt{d}}\right), \qquad \sum_{j=1}^{N} A_{ij}^{(\ell,b)} = 1, \tag{13}$$

where q and k are the query and key projections and d is the head dimension. For a query token i we measure how much of its attention is directed to the visual stream by summing over visual keys and averaging over heads,

$$a_{i\to\mathcal{I}}^{(\ell)} = \frac{1}{H} \sum_{b=1}^{H} \sum_{j\in\mathcal{I}} A_{ij}^{(\ell,b)}, \tag{14}$$

which we refer to as the visual attention ratio of token i at layer ℓ. Aggregating $a_{i\to\mathcal{I}}^{(\ell)}$ over the textual query tokens gives a scalar summary of how strongly the model attends to the image while producing its response. A well-documented empirical observation is that this ratio is high in the first few layers but decays sharply with depth, so that in deep layers the model attends almost entirely to textual tokens [3].

Table 4: Controlled textual-interference results on the constructed GTA subset. All values are percentages. Random-1 results are averaged over random draws; cases without an eligible deletion remain unchanged. SPARE-V5 is evaluated as an automatic procedure rather than with a manually specified deletion mask.

Model	Method	Acc. ↑	SCR ↓
Qwen3-VL-8B	Full Trace	63.64	36.36
	Random-1	72.73	27.27
	SPARE-V5	**90.91**	**9.09**
Qwen3-VL-30B-A3B	Full Trace	54.55	45.45
	Random-1	72.73	22.73
	SPARE-V5	**81.82**	**18.18**

B Supplementary Analyses of Textual Interference

B.1 Controlled Diagnostic of Textual Interference

Relation to the main experiments. The main experiments evaluate the complete SPARE pipeline across multiple MLLM backbones and full benchmarks. Complementary to those end-to-end results and the attention analysis, we conduct an additional controlled diagnostic to test whether pruning stale reasoning makes the model less likely to follow an obsolete textual hypothesis when later visual or tool evidence supports a different answer. This diagnostic is separate from the aggregate benchmark evaluation.

Controlled construction. We construct a subset from GTA containing audited cases in which an early reasoning segment expresses a plausible but incorrect answer and a later tool observation provides corrective evidence. Across conditions, the original question, image, tool calls, action blocks, and tool observations are held fixed; only the retained reasoning history changes.

We compare three conditions. **Full Trace** keeps the complete history, including the stale hypothesis. **Random-1** removes one eligible reasoning segment at random, providing a control for context shortening without targeted selection. **SPARE-V5** automatically applies its pruning procedure to the eligible reasoning history while preserving the tool interaction and visual evidence. Every audited conflict case is retained in the evaluation, including cases in which pruning does not activate and the context therefore remains unchanged.

Metrics. We report forced-choice accuracy (Acc.), which measures how often the model selects the answer supported by the later evidence, and stale-copy rate (SCR), which measures how often the final prediction instead repeats the obsolete textual hypothesis. Formally, for method m,

$$\mathrm{Acc}_m = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\hat{y}_i^m = y_i\right], \tag{15}$$

$$\mathrm{SCR}_m = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\hat{y}_i^m = s_i\right], \tag{16}$$

where y_i and s_i denote the ground-truth and stale answers, respectively. A lower SCR indicates that the final decision is less likely to follow obsolete textual reasoning when it conflicts with later evidence.

Results. For Qwen3-VL-8B, SPARE-V5 improves forced-choice accuracy from 63.64% to 90.91% and reduces stale-answer copying from 36.36% to 9.09%. For Qwen3-VL-30B-A3B, accuracy increases from 54.55% to 81.82%, while SCR decreases from 45.45% to 18.18%. Thus, across both backbones, SPARE-V5 improves accuracy and reduces stale copying by 27.27 percentage points.

Random-1 produces smaller improvements than SPARE-V5 for both backbones. This comparison indicates that the benefit cannot be explained solely by shortening the reasoning history: selecting which reasoning content to remove is important for reducing interference from stale textual hypotheses. Free-form answer accuracy remains unchanged between Full Trace and SPARE-V5, suggesting that

the observed effect reflects reduced reliance on conflicting textual history rather than a general change in task-solving ability.

Scope. This deliberately controlled diagnostic is intended as mechanism-level support for the textual-debt motivation, rather than as a separate benchmark result. It shows that the automatic SPARE-V5 procedure can reduce interference from stale reasoning while retaining the later visual and tool evidence. Because the diagnostic isolates a specific form of text–evidence conflict, it should not be interpreted as a general comparison between SPARE-V5 and every possible reasoning-compression strategy.

B.2 Attention Visualization After Pruning

To further examine whether pruning reduces textual dominance and restores reliance on visual evidence, we visualize per-layer text-to-image attention averaged over 10 tool-use trajectories. After applying SPARE, attention to image tokens increases across nearly all layers. This supports our motivation: redundant accumulated text can dominate the context and distract the model from visual evidence, while pruning it reallocates computation back to the image. The cross-layer attention shift provides additional evidence that suppressing textual dominance restores the role of visual evidence in downstream reasoning.

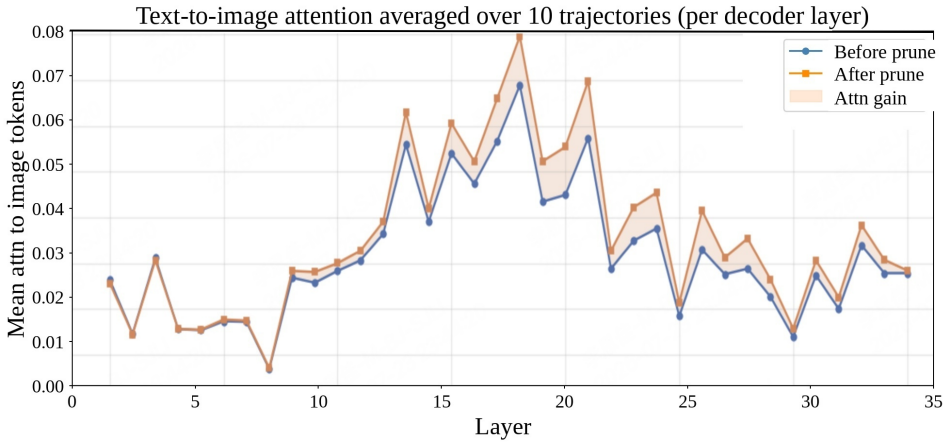


Figure 2: Text-to-image attention across decoder layers, averaged over 10 tool-use trajectories. Blue and orange denote attention before and after pruning, respectively, and the shaded region shows the gain. Pruning summary-covered reasoning consistently increases attention to image tokens, indicating stronger reliance on visual evidence.

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    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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